

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Final Examination of Semester 1
Subject: 031021321 Electrical Drawing I
Date: 7 October 2024
Year: 2024
Section: 5-6
Time: 08.00-10.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. There will be 7 pages (including this page) and they are worth 40 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on the answer sheet.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators cannot be used in this test.

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Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

This exam paper is part of a set used to assess student abilities as follows:

		Q1	Q2
CLO 1	Draw symbols for IEC standard correctly	√	√
CLO 2	Draw interior electrical equipment correctly	√	√
CLO 3	Draw sub distribution board for home correctly	√	√
CLO 4	Draw circuit for measuring instruments correctly	√	√

1. The floor plan of building “A” is shown in Figure 1. Please design the circuits, then complete one-line and control diagrams. (20 points)

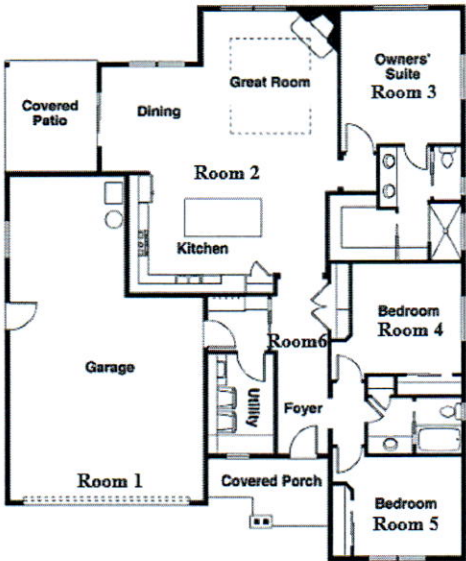


Figure 1

The floor plan is as follows.

Load Center	Room	Switch	Lamp	Plug Socket
1LC	1	S1, S2 & S5	L1 and E11 with series connection	-
1LC	1	S6	E1 and E12 with parallel connection	-
2LC	2	S3 & S4	L2 and E4 with parallel connection	-
2LC	2	S7	L3, E6 and E13 with series connection	-
3LC	3	S11	E5 and E8	-
4LC	4	S8	E2 and E7 with parallel connection	-
4LC	5	S9	E3 and E9 with parallel connection	-
4LC	6	S10	E10	-
5LC	1 & 2	-	-	2

Symbol of One Line Diagram	Building “A”
	E1-E3
	E4-E5
	E6-E7
	E8-E13
	L1-L3
	S1-S4
	S5
	S6-S10
	S11

2. The floor plan of building “B” is shown in Figure 2. Please design the circuits, then complete one-line and control diagrams. (20 points)

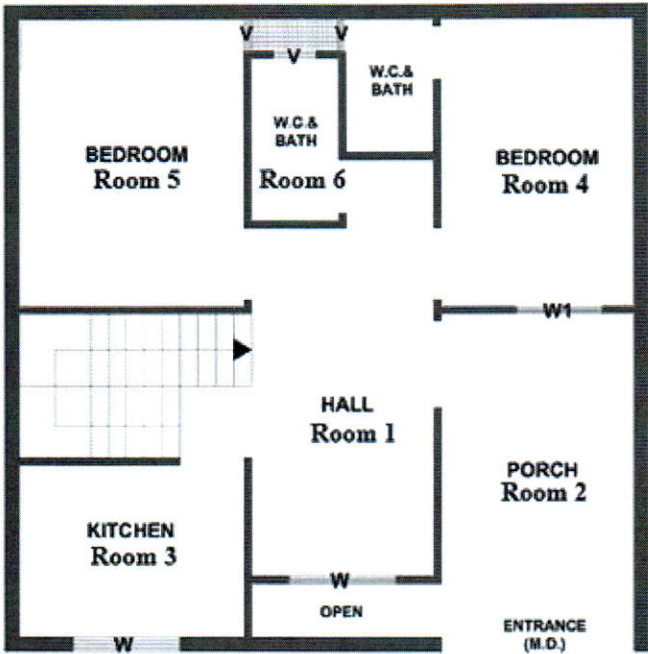


Figure 2

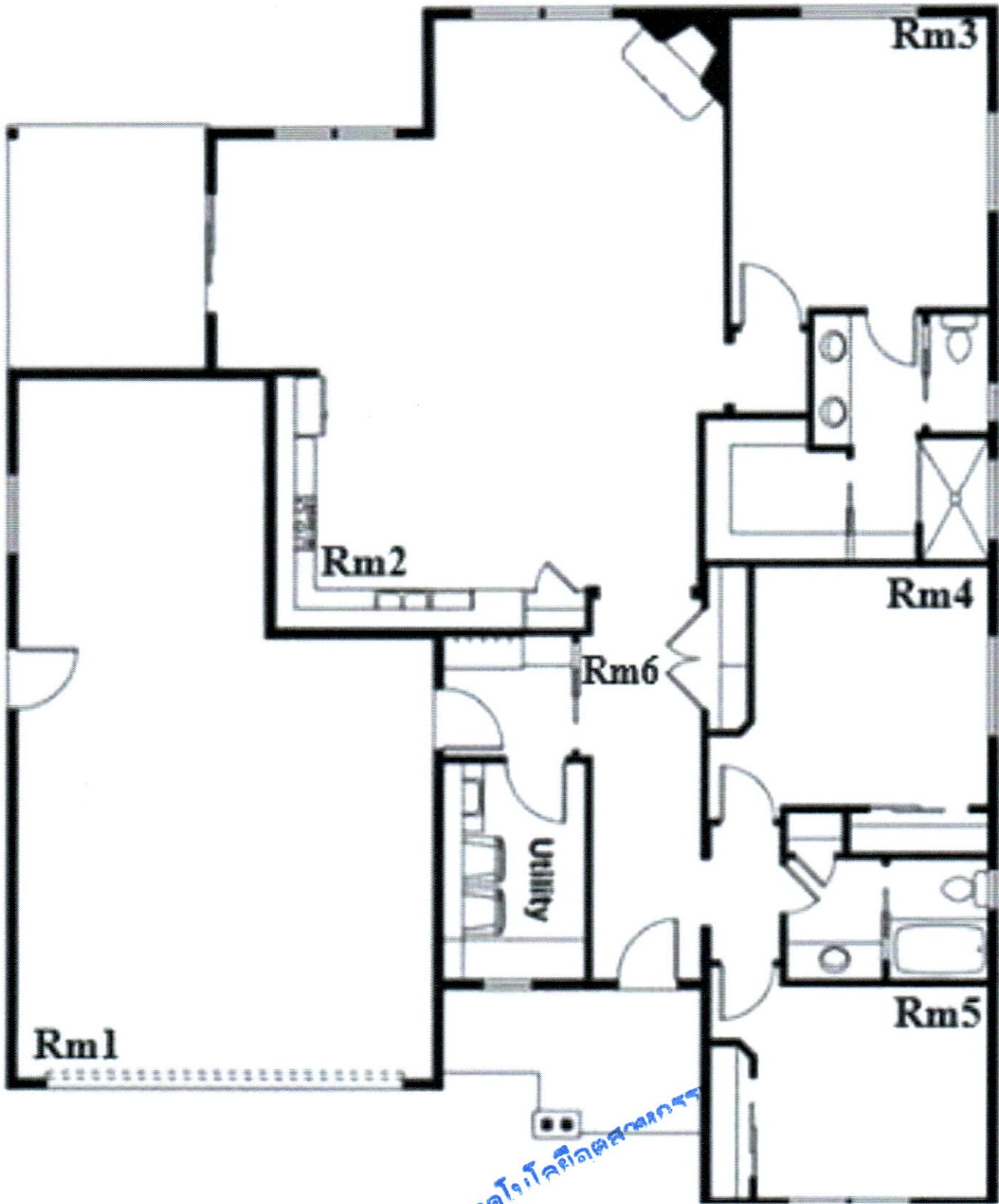
The floor plan is as follows.

Load Center	Room	Switch	Lamp	Plug Socket
1LC	1	S1	L1 & E1 with series connection	-
1LC	1	S4	E5	-
1LC	1	S7 & S8	L2 & E7 with series connection	-
2LC	2	S2	E8	-
3LC	3	S5	L3 & E10	-
3LC	4	S9 & S10	E9 & E11 with series connection	-
4LC	5	S3	E2, E3 & E4 with parallel connection	-
4LC	6	S6	E6	-
5LC	4 & 5	-	-	2

Symbol of One Line Diagram	Building “A”
	E1-E4
	E5-E6
	E7-E9
	E10-E11
	L1-L3
	S1-S3
	S4-S6
	S7-S10

Answer Sheet

One-line diagram of building “A” (10 points)



Name:

Student ID:

Score:

Answer Sheet

#1

Control diagram of building “A” (10 points)

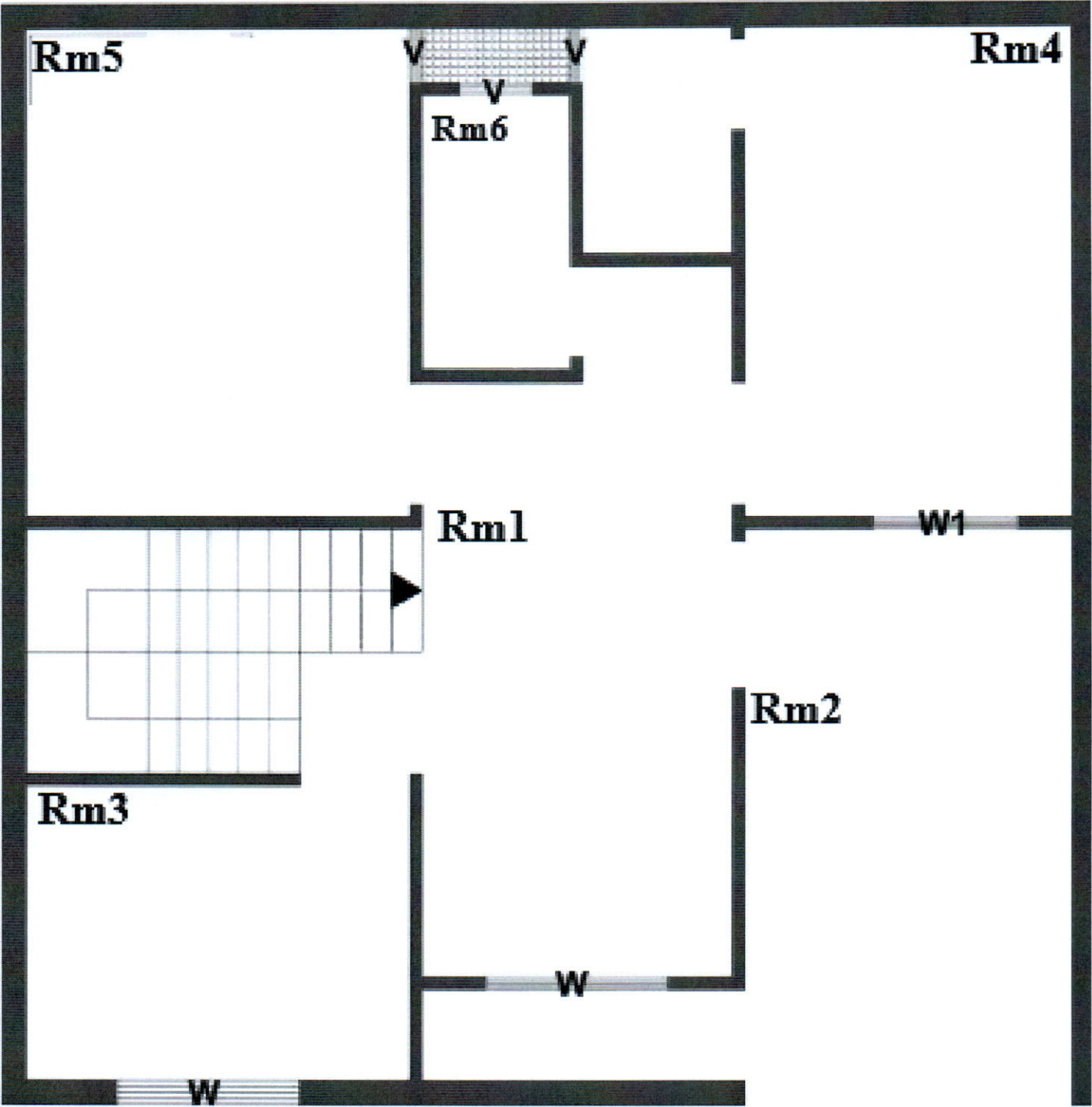
L1

N

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Name:	Score:	Answer Sheet #2
Student ID:		

One-line diagram of building “B” (10 points)



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Name:	Score:	Answer Sheet
Student ID:		#3

Control diagram of building “B” (10 points)

L1

N

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Name:	Score:	Answer Sheet
Student ID:		#4